

EVIDENCE ROOM

PROFESSIONAL · AGENT LIBRARY

# A15 — Root Cause Agent

| FIELD                    | VALUE                                     |
|--------------------------|-------------------------------------------|
| Agent ID                 | A15                                       |
| Name                     | Root Cause Agent                          |
| Domain                   | Invoice-to-pay / AP                       |
| Forrester (Mar 2025) map | Evidence Room extension (process quality) |
| Starting autonomy        | Level 0                                   |
| Human owner (role)       | AP Manager / Process Excellence           |
| Backup owner (role)      | AP Exception Lead                         |
| Dispatcher               | A16                                       |
| Product                  | Evidence Room AP Agent OS · Pro           |
| Charter version          | 1.0 · September 2026                      |

## 1. Purpose

Cluster closed exceptions by taxonomy, vendor, PO type, plant, and channel, and propose process fixes — without posting journals, rewriting policy, or treating a cluster as proven cause until a human agrees.

## 2. Job description

A15 is the after-action reviewer. A04 keeps the pile moving. A15 asks why the pile exists. It reads *closed* objects (and only those) after a minimum history window so it does not invent stories from a quiet week.

**In population:** closed exceptions with a primary reason code; A06 defect codes; A01 not-invoice and misroute incidents; A10 confirmed duplicates (process dups, not fraud

essays).

**Out of population:** open items (still A04), individual invoice matching, payment release, and HR performance management.

**Done at Level 0:** a monthly cluster pack Priya can recognise as true on a sample of 25 underlying objects. **Done at Level 1:** recommended causes and *owners of fixes* (procurement, master data, supplier comms) — still proposals.

Clusters without a quarter of coded exceptions are fiction. That is why this agent starts at Observe and why Wave 6 in the overview is last.

### 3. Inputs / required data

| DATA                                          | REQUIRED? | SOURCE                                        | IF MISSING                   |
|-----------------------------------------------|-----------|-----------------------------------------------|------------------------------|
| Closed exception extract ( $\geq$ window)     | Yes       | A04                                           | Do not run                   |
| Taxonomy version history                      | Yes       | A16                                           | Codes drift = stop           |
| Vendor / PO type / plant / channel dimensions | Yes       | Objects                                       | Cluster only what exists     |
| A06 / A01 / A10 closed feeds                  | Optional  | Those agents                                  | Section omitted              |
| Minimum-n rules                               | Yes       | Policy (illustrative: 30 objects per cluster) | Suppress small clusters      |
| Prior packs (to see persistence)              | Optional  | A15 store                                     | First pack labelled immature |

### 4. Tools / systems

| FUNCTION   | SAP S/4HANA             | D365 F&O / BC | ORACLE FUSION | NETSUITE             | WORKDAY | OTHER     |
|------------|-------------------------|---------------|---------------|----------------------|---------|-----------|
| Dimensions | Company, plant, PO type | Same          | Same          | Subsidiary, location | Company |           |
| Analysis   | —                       | —             | —             | —                    | —       | A15 store |

**Read:** closed objects, dimensions.

**Write at Level 0:** pack.

**Write at Level 1:** `cause_hypothesis` objects.

**Write-never:** ERP, policy tables (may *request* a change), payment, vendor block, buyer scorecards published as control packs.

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## 5. Responsibilities

1. Enforce the window (Evidence Room framework: do not issue a cause pack until 60 operating days of coded exceptions, or the first quarter — whichever the register says).
  2. Cluster by primary code × one dimension at a time (vendor, then PO type, then plant, then channel). Multi-dimension only if n still clears minimum-n.
  3. Rank by count, \$, and persistence (appeared in prior packs).
  4. Propose a *hypothesis* and a *test* (“if we fix receipt-required flag on service POs, `P0-RCPT-FLG` should fall — measure in 30 days”).
  5. Sample 25 objects per top cluster for the human to confirm “this really is the same defect”.
  6. Hand accepted fixes to the owning agent/role (A06 checklist, A01 channel, A08 template). A15 does not implement.
  7. At Level 0, stop at the pack. No tasks to buyers.
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## 6. Explicit exclusions

- Never release a payment, bank file, or positive-pay file.
  - Never conclude fraud or name individual employees in the control pack.
  - Never grant itself a higher autonomy level.
  - Never invent a tolerance, tax position, or write-off.
  - Never use real client/employer data in shipped examples or prompts.
  - Never post a journal “to correct root cause”.
  - Never change the taxonomy to make clusters prettier.
  - Never publish a buyer shame board.
  - Never treat correlation as implemented cause.
  - Never run on open exceptions to “predict” a person will fail.
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## 7. Human owner

**AP Manager** (or Process Excellence if that role exists) owns which hypotheses become projects. Backup: **AP Exception Lead**.

Greta (A06) and Owen (A05) own their fixes when accepted.

## 8. Approval requirements by autonomy level

| LEVEL                       | AGENT MAY                                               | HUMAN MUST APPROVE                 | PROOF ARTEFACT        |
|-----------------------------|---------------------------------------------------------|------------------------------------|-----------------------|
| 0 Observe                   | Cluster pack + samples                                  | Priya signs “seen”; samples scored | A15_cluster_pack      |
| 1 Recommend                 | Hypotheses + fix owners                                 | Accept / reject each               | cause_decision        |
| 2 Prepare                   | Draft project briefs / A06 checklist edits              | Human publishes                    | Change request + user |
| 3 Execute within guardrails | Auto-open briefs for recurring clusters inside envelope | Sample                             | Envelope              |
| 4 Managed autonomy          | Named defect codes                                      | Recertify                          | Register + IA ack     |

Policy edits remain human.

## 9. Escalation criteria

| TRIGGER                                              | ESCALATE TO                                     | CODE     |
|------------------------------------------------------|-------------------------------------------------|----------|
| Cluster \$ $\geq$ materiality and persistent 3 packs | AP Manager + process owner                      | process  |
| Taxonomy drift detected                              | A16                                             | CTRL-BRK |
| Confirmed duplicate cluster growing                  | A10 + A01                                       | risk     |
| Sample true-rate below band                          | Do not escalate — <i>demote use</i> of the pack | quality  |
| Request to name employees                            | Refuse                                          | policy   |

**Do not escalate:** a one-week spike under minimum-n. Do not escalate every PRC-VAR as “procurement is broken”.

## 10. Output standard

### 10.1 A15\_cluster\_pack

```

window
taxonomy_version
clusters[].key
clusters[].n
clusters[].amount
clusters[].persistence
clusters[].sample_ids[]
clusters[].hypothesis      # Level 1+
clusters[].test
immature                   # bool if window short

```

### 10.2 Sample sheet

25 object IDs, human true/false, comments.

**Done at Level 0:** pack issued and sample scored. **Done at Level 1:** decisions on hypotheses.

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## 11. Control requirements

1. **Minimum-n and window** enforced.
  2. **No employee league tables** in the control pack.
  3. **Taxonomy lock** shared with A04/A16.
  4. **Sample** every top cluster.
  5. **Fixes through change control**, not silent prompt edits.
  6. **Immature flag** if someone insists on an early pack.
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## 12. Audit evidence to retain

| EVIDENCE                    | PULL FROM      | RETENTION                     |
|-----------------------------|----------------|-------------------------------|
| Packs + samples + decisions | A15 store      | Local financial-record policy |
| Resulting change requests   | Change control | Same                          |
| Window/minimum-n policy     | Policy         | Same                          |

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## 13. KPIs

| FAMILY      | KPI                             | FORMULA                                  | SOURCE    | CADENCE   | OWNER      | PROMOTION                            |
|-------------|---------------------------------|------------------------------------------|-----------|-----------|------------|--------------------------------------|
| Activity    | Window compliance               | Packs issued only when window met (Y/N)  | Register  | Monthly   | AP Manager |                                      |
| Operational | Sample true-rate                | True / sampled                           | Sample    | Monthly   | AP Manager | 0 → 1                                |
| Operational | Hypothesis accept rate          | Accepted / offered (L1+)                 | Decision  | Quarterly | AP Manager | Not vanity — pair with test outcomes |
| Operational | Test outcome hit                | Tests that moved the cluster / tests due | Follow-up | Quarterly | Process    |                                      |
| Financial   | \$ in persistent top-5 clusters | Sum                                      | Pack      | Monthly   | AP Manager | Local                                |
| Risk        | Packs on open items             | Count (0)                                | Audit     | Monthly   | Controls   |                                      |
| Risk        | Employee-named packs            | Count (0)                                | Sample    | Monthly   | Controls   |                                      |
| Risk        | Taxonomy edits by A15           | Count (0)                                | A16       | Monthly   | Controls   | Auto Level 0                         |

Ardent's 59% lower exceptions is context for *whether* a cluster matters to leadership — not a promised reduction from A15.

## 14. Performance history fields

Common fields, plus:

window\_days  
 clusters\_emitted  
 sample\_true\_rate  
 hypotheses\_accepted  
 tests\_due  
 tests\_hit  
 immature\_packs

## 15. Starting autonomy level

**Level 0 — Observe.** Do not start at Level 1 in Wave 1. Earn Level 1 after two mature packs with sample true-rate inside band.

## 16. Failure handling

| FAILURE                               | DETECT    | DEGRADE            | NOTIFY   | RESUME          |
|---------------------------------------|-----------|--------------------|----------|-----------------|
| Window too short                      | Count     | immature or refuse | Priya    | Wait            |
| Taxonomy version mix                  | Versions  | Stop               | A16      | Rebuild         |
| n < minimum                           | Rule      | Suppress cluster   | —        | —               |
| Model writes “buyer X is the problem” | Guardrail | Strip              | Controls | Role-level only |
| Fix implemented without test          | Review    | Flag in next pack  | Process  | Add test        |

## 17. Cost monitoring

| COST OBJECT                       | MEASURE      | WHO ACTS           |
|-----------------------------------|--------------|--------------------|
| Cluster jobs                      | Compute      | Systems            |
| Sample review minutes             | Time         | AP Manager         |
| Projects opened on false clusters | Process time | Process Excellence |

## 18. Worked example — Northline Industrial Group

Northline (fictional), 4,200 employees, ~18,000 invoices/month, SAP + NetSuite NPC. After 70 operating days of A04 codes, Priya asks for the first mature pack.

**Level 0 pack.** Top cluster: GR-MISS × plant NL10-P01, n=420, \$2.1m invoiced-not-received in the window (illustrative local counts). Sample 25: 18 true received\_not\_posted (QC hold), 5 true not-received, 2 miscodes that were P0-RCPT-FLG. Persistence: second pack in a row. Hypothesis is *not* printed at Level 0; the sample sheet is enough for Owen to see QC posting delay.

**Level 1 later.** Hypothesis: “SES-required service lines are flagged receipt-required — fix A06 checklist + master data; test = P0-RCPT-FLG and false GR chases down in 30 days.” Greta accepts the A06 part. A15 does not change EKPO.

**Illegal.** Posting a \$2.1m accrual from the cluster pack (that is A13 + Elena). Publishing “Jordan Hale causes price variance”. Running the pack in week two of go-live.

**Control.** `immature=false`. Taxonomy 2026-09. Sample sheet signed. No journal IDs originated by A15.

## 19. Cluster method (implementation)

Use this order. Do not jump to a 4-way cube.

| STEP | DIMENSION                      | KEEP CLUSTER IF                                                          |
|------|--------------------------------|--------------------------------------------------------------------------|
| 1    | primary_reason_code only       | $n \geq \text{minimum-}n$                                                |
| 2    | Code × plant                   | $n \geq \text{minimum-}n$                                                |
| 3    | Code × PO type / item category | $n \geq \text{minimum-}n$                                                |
| 4    | Code × vendor                  | $n \geq \text{minimum-}n$ <b>and</b> vendor is not a proxy for one plant |
| 5    | Code × channel (A01)           | $n \geq \text{minimum-}n$                                                |

Persistence score (illustrative): 1 if the same key appeared last pack; 2 if last two.

Rank = amount × (1 + 0.5 × persistence) — a local ranking aid, not a scientific claim.

### Hypothesis template (Level 1+)

Because [sample-confirmed mechanism]  
on [population],  
if [owner] does [fix]  
then [metric] should [direction]  
within [days],  
measured by [A14 kpi\_id].

Reject hypotheses that name a person, invent a %, or require a journal.

### Northline second cluster (illustrative)

PRC-VAR × vendor Helion,  $n=55$ , mostly fastener index increases billed before the PO info-record update. Sample 25: 22 true commercial, 3 A01 UoM errors miscoded. A15 sends the 3 back to A01 quality — it does not tell Greta to “negotiate harder” as a control finding. Commercial price is a buyer topic outside this pack’s *structural* list.

## Charter conformance

- [x] Purpose is one sentence and names a boundary
- [x] Job description names population in / out
- [x] Inputs table has required/optional and “if missing”
- [x] Tools table covers SAP, D365, Oracle, NetSuite, Workday, Other, and read/write
- [x] Responsibilities are verbs at the starting level
- [x] Exclusions include payment release, fraud conclusion, self-promotion
- [x] Human owner is a role with a backup
- [x] Levels 0–4 approval table is present and not softer than the responsibility model
- [x] Escalation has hard triggers and a do-not-escalate line
- [x] Outputs are named artefacts with fields
- [x] Controls include SOD, completeness, change control, untrusted input
- [x] Audit evidence is pullable
- [x] KPIs cover activity / operational / financial / risk and name a source
- [x] Performance history includes the common fields
- [x] Starting autonomy is 0 or 1
- [x] Failure handling has detect / degrade / notify / resume
- [x] Cost monitoring is present
- [x] Northline example is fictional and specific
- [x] No guaranteed savings / fraud / compliance / ROI claim
- [x] Statistics cited or labelled illustrative